DS6410 — Advanced Machine Learning

Dimension Reduction and Latent Variables

Reference: Bishop, C.M. (2006). *Pattern Recognition and Machine Learning*, Chapter 12

What We Will Cover Today

This lecture is structured in two parts. The first part is a student-led mini-presentation reviewing the conceptual foundations of unsupervised learning. The second part is an instructor-led lecture covering the mathematical formulations of PCA, its extensions, and an introduction to ICA.

PART	TOPIC	FORMAT
Part I	Unsupervised Learning: Concepts and Context	*Student Mini-Presentation (~10 min)*
Part II	PCA: Maximum Variance & Minimum Error Formulations	*Instructor Lecture*
Part III	High-Dimensional PCA, Sparse PCA, Probabilistic PCA	*Instructor Lecture*
Part IV	Kernel PCA — Nonlinear Dimensionality Reduction	*Instructor Lecture*
Part V	Independent Component Analysis (ICA)	*Instructor Lecture (time permitting)*

Supervised vs. Unsupervised Learning: What Changes?

Supervised Learning

Every training example consists of an input $\mathbf{x}$ paired with a corresponding label or target y. The learning objective is to approximate the mapping $f: x \rightarrow y$, typically by minimizing a loss function that directly measures prediction error. The availability of labels provides a clear, externally defined signal of correctness.

Unsupervised Learning

No labels are provided. The algorithm receives only the input data $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ and must discover structure, patterns, or compact representations entirely from the data's intrinsic geometry and statistics. There is no external oracle to define what "correct" means.

DIMENSION	SUPERVISED LEARNING	UNSUPERVISED LEARNING
Learning Objective	Minimize prediction error w.r.t. known targets	Discover intrinsic structure, clusters, or low-dimensional representations
Data Required	Labeled pairs $(\mathbf{x}_i, y_i)$	Unlabeled observations $\{\mathbf{x}_i\}$ only
Evaluation	Quantitative metrics (accuracy, MSE, AUC) on held-out labeled data	Indirect: reconstruction error, cluster cohesion, downstream task performance
Feedback Signal	Direct — loss gradient w.r.t. labels	Indirect — internal criteria (variance explained, likelihood)

When Supervised Learning Is Insufficient or Infeasible

Three canonical situations where unsupervised learning is the only viable approach:

1. Labels are unavailable or expensive

Annotating large datasets requires domain expertise, time, and cost. In genomics, labeling millions of gene expression profiles with clinical phenotypes is not always feasible, yet the data contains rich biological structure.

2. The goal is exploratory

When a researcher does not yet know what to predict, unsupervised methods help reveal the latent structure of the data — identifying natural groupings, detecting anomalies, or finding low-dimensional manifolds before any hypothesis is formed.

3. High-dimensional input space

In text, images, or genomic data, the raw feature space may have tens of thousands of dimensions. Unsupervised dimensionality reduction (e.g., PCA) is used as a preprocessing step to extract a compact, informative representation.

CONCRETE EXAMPLE

Single-cell RNA sequencing (scRNA-seq)

Each cell is described by the expression levels of ~20,000 genes. No labels exist for cell type at the time of measurement.

Researchers apply **PCA** and clustering (e.g., k-means or UMAP) to identify distinct cell populations — an entirely unsupervised pipeline.

PCA: What Problem Does It Solve?

THE PROBLEM

High-dimensional data is difficult to visualize, computationally expensive to process, and often contains redundancy from correlated features.

> Given N observations in D-dimensional space, find an M-dimensional linear subspace ($M << D$) that retains as much variance as possible.

THE MAIN IDEA

PCA finds a small number of orthogonal directions, called **principal components**, that capture the maximum variance in the data.

The first principal component u_1 is the direction of maximum variance; subsequent components are orthogonal and capture the next-largest variance.

KEY ASSUMPTIONS

- **Variance = Information:** The meaningful structure in the data lies along directions of high variance.
- **Linearity:** The relationships among variables are approximately linear.
- **Gaussianity:** The data is approximately Gaussian, or at least well-characterized by its second-order statistics (mean and covariance).
- **Scale:** Features are measured on a comparable scale, or the data has been standardized prior to analysis.

PCA: Strengths, Limitations, and Failure Modes

WHEN PCA IS MOST USEFUL

- Data lies near a low-dimensional **linear subspace** (e.g., face images under varying illumination).
- The goal is **dimensionality reduction** prior to regression, classification, or visualization.
- Features are highly correlated and redundant (e.g., spectroscopic measurements across many wavelengths).
- The dataset is large and high-dimensional, making direct modeling intractable.

✓ Where PCA is Helpful

Face Recognition (Eigenfaces): Thousands of pixel-valued face images are projected onto the top principal components. A small number of components (~50–100) captures most of the variance and enables efficient recognition.

✗ Where PCA Fails

The Swiss Roll: Consider data distributed along a 2D manifold embedded nonlinearly in 3D space. PCA, being a linear method, cannot "unroll" this manifold. It projects the data onto a 2D plane, mixing points that are geometrically far apart.

The Core Challenges of Unsupervised Learning

Unsupervised learning is fundamentally harder than supervised learning, for several interconnected reasons.

01 No ground truth for evaluation

Without labels, there is no universally agreed-upon criterion for success. Metrics such as reconstruction error, silhouette score, or explained variance are proxies — they measure internal consistency, not true correctness.

02 The curse of dimensionality

In high dimensions, distances become increasingly uniform, making it difficult to identify meaningful structure. Density estimation and clustering become unreliable without dimensionality reduction.

03 Model selection & tuning

Choosing the number of principal components, clusters, or latent dimensions is non-trivial. Cross-validation requires a supervised signal; unsupervised analogues (e.g., scree plots, BIC) are heuristic.

04 Identifiability

Many unsupervised models are not identifiable — multiple parameter settings yield the same likelihood (e.g., rotational invariance in PCA's latent space). This is a fundamental statistical issue, not a computational one.

05 Interpretability

Latent dimensions discovered by unsupervised methods may not correspond to human-interpretable concepts, making it difficult to validate or communicate findings.

From Intuition to Mathematics

The central theme of this chapter is that PCA can be derived from multiple, complementary perspectives — each revealing a different facet of the same underlying structure. We will cover:

1. **Maximum Variance Formulation**
   PCA as variance maximization (Hotelling, 1933)
2. **Minimum Error Formulation**
   PCA as optimal linear reconstruction (Pearson, 1901)
3. **High-Dimensional PCA**
   Efficient computation when $D \gg N$
4. **Sparse PCA**
   Regularized PCA for interpretable components
5. **Probabilistic PCA**
   A generative model perspective (Tipping & Bishop, 1999)
6. **Kernel PCA**
   Nonlinear extension via the kernel trick
7. **ICA**
   Beyond second-order statistics

Setup: Data, Centering, and the Covariance Matrix

Let $\{\mathbf{x}_n\}_{n=1}^N$ be a dataset of N observations, each in $\mathbb{R}^D$. We assume the data has been **mean-centered**:

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n, \qquad \tilde{\mathbf{x}}_n = \mathbf{x}_n - \bar{\mathbf{x}}$$

The **sample covariance matrix** is:

$$\mathbf{S} = \frac{1}{N}\sum_{n=1}^{N}\tilde{\mathbf{x}}_n\tilde{\mathbf{x}}_n^\top \in \mathbb{R}^{DtimesD}$$

This is a symmetric, positive semi-definite matrix.

Its **eigendecomposition** is:

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$$

where $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_D]$ is an orthogonal matrix of eigenvectors and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_D)$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$.

THE GOAL OF PCA

Find an M-dimensional linear subspace ($M < D$) that best represents the data.

Maximum Variance Formulation of PCA

We seek a unit vector $u_1 \in \mathbb{R}^D$ ($||u_1|| = 1$) such that the **variance of the projected data** is maximized.

The projection of $\tilde{x}_n$ onto u_1 is the scalar $z_{n1} = u_1^T \tilde{x}_n$. The variance of these projections is:

$$\mathrm{Var}(z_1) = (1/N) \Sigma (u_1^T \tilde{x}_n)^2 = u_1^T S u_1$$

Optimization problem:

$$\max u_1^T S u_1 \quad \text{subject to} \quad u_1^T u_1 = 1$$

Using a Lagrange multiplier λ_1, the stationarity condition gives:

$$S u_1 = \lambda_1 u_1$$

> **Key result:** u_1 must be an **eigenvector** of S, and the maximized variance equals the corresponding **eigenvalue** λ_1. To maximize variance, choose the eigenvector for the **largest eigenvalue**.

By induction, u_m is the eigenvector of S for the m-th largest eigenvalue λ_m, orthogonal to previous components.

PCA via Singular Value Decomposition (SVD)

THE SVD OF THE DATA MATRIX

In practice, PCA is rarely computed by explicitly forming the covariance matrix. Instead, we use the **Singular Value Decomposition (SVD)** of the centered data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

- $\mathbf{U} \in \mathbb{R}^{N \times N}$: Orthogonal matrix of left singular vectors.
- $\mathbf{\Sigma} \in \mathbb{R}^{N \times D}$: Diagonal matrix of singular values σ_i.
- $\mathbf{V} \in \mathbb{R}^{D \times D}$: Orthogonal matrix of right singular vectors.

CONNECTION TO COVARIANCE

The sample covariance matrix is $\mathbf{S} = \frac{1}{N}\mathbf{X}^\top\mathbf{X}$. Substituting the SVD of $\mathbf{X}$:

$$\mathbf{S} = \frac{1}{N}(\mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top)(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) = \mathbf{V}\left(\frac{\mathbf{\Sigma}^2}{N}\right)\mathbf{V}^\top$$

This is exactly the eigendecomposition of $\mathbf{S}$.

Key Takeaways:

- The right singular vectors $\mathbf{V}$ are the **principal directions** (eigenvectors of $\mathbf{S}$).
- The eigenvalues are related to the singular values by $\lambda_i = \sigma_i^2/N$.
- SVD is more **numerically stable** than forming $\mathbf{X}^\top\mathbf{X}$ explicitly.

Minimum Error Formulation of PCA

PCA finds the linear subspace that minimizes the **reconstruction error** when data are projected and reconstructed.

Represent each data point by an **M**-dimensional code $\mathbf{z}_n \in \mathbb{R}^M$ and reconstruct via:

$$\tilde{\mathbf{x}}_n \approx \hat{\mathbf{x}}_n = \sum_{m=1}^{M} z_{nm}\mathbf{u}_m + \sum_{m=M+1}^{D} b_m \mathbf{u}_m$$

Here $\{b_m\}$ are constants for discarded dimensions. The **distortion measure** (mean squared reconstruction error) is:

$$J = \frac{1}{N}\sum_{n=1}^{N} \left\|\tilde{\mathbf{x}}_n - \hat{\mathbf{x}}_n\right\|^2$$

Key result: Minimizing J over orthonormal bases $\{\mathbf{u}_m\}$ and constants $\{b_m\}$ yields the same solution as the maximum-variance view: the optimal basis vectors are the eigenvectors of $\mathbf{S}$.

The minimized distortion equals the **total variance in the discarded dimensions**:

$$J_{\min} = \sum_{m=M+1}^{D} \lambda_m$$

The two formulations — maximize retained variance, minimize discarded variance — are equivalent since total variance is fixed: $\sum_{m=1}^{D} \lambda_m = \mathrm{tr}(\mathbf{S}) = \mathrm{const}$.

How Many Components? The Scree Plot and Explained Variance

A central practical question in PCA is: how many principal components M should we retain?

PROPORTION OF VARIANCE EXPLAINED

The proportion of variance explained (PVE) by the first M components is the ratio of the sum of the first M eigenvalues to the sum of all D eigenvalues:

$$\mathrm{PVE}(M) = \frac{\sum_{m=1}^{M} \lambda_m}{\sum_{d=1}^{D} \lambda_d}$$

SELECTION HEURISTICS

- **Variance Threshold:** Retain components that explain a target percentage (e.g., 90–95%) of the total variance.
- **Scree Plot:** Plot λ_m vs. m and look for an "elbow" where the eigenvalues begin to level off.
- **Probabilistic Framework:** In PPCA (Tipping & Bishop, 1999), the optimal M can be selected by maximizing the marginal likelihood, providing a principled Bayesian model selection criterion.

PCA in High Dimensions: When D ≫ N

In many modern applications — genomics, text, images — the dimensionality D far exceeds the number of observations N. Computing the $D \times D$ covariance matrix $\mathbf{S}$ is computationally expensive ($O(D^2N)$ to form, $O(D^3)$ to diagonalize), and $\mathbf{S}$ has rank at most $N-1$, so at most $N-1$ non-zero eigenvalues exist.

The dual trick (Bishop §12.1.4): Define the $N \times N$ matrix:

$$\mathbf{X}\mathbf{X}^\top \in \mathbb{R}^{N\times N}, \quad \text{where } \mathbf{X} = [\tilde{\mathbf{x}}_1, \ldots, \tilde{\mathbf{x}}_N]^\top$$

If $\mathbf{v}_i$ is an eigenvector of $\mathbf{X}\mathbf{X}^\top$ with eigenvalue λ_i, then $\mathbf{u}_i = \frac{1}{\sqrt{N\lambda_i}}\mathbf{X}^\top\mathbf{v}_i$ is the corresponding eigenvector of $\mathbf{S} = \frac{1}{N}\mathbf{X}^\top\mathbf{X}$.

This reduces the eigenvalue problem from $D \times D$ to $N \times N$, which is far more tractable when $D \gg N$.

SCENARIO	COVARIANCE MATRIX	EIGENVALUE PROBLEM	COST
$N \gg D$	$D \times D$	Standard	$O(D^3)$
$D \gg N$	$N \times N$ (dual)	Dual trick	$O(N^3)$

Sparse PCA: Mathematical Formulation

THE ELASTIC NET FORMULATION

Standard PCA components are linear combinations of all D features. Zou, Hastie & Tibshirani (2006) formulated PCA as a regression-type optimization problem and added an **Elastic Net** penalty to induce sparsity.

For the first M components, we minimize over $\mathbf{A} \in \mathbb{R}^{D \times M}$ and $\mathbf{B} \in \mathbb{R}^{D \times M}$:

$$\min_{\mathbf{A},\mathbf{B}} \sum_{i=1}^{N} \left\| \mathbf{x}_i - \mathbf{A}\mathbf{B}^\top \mathbf{x}_i \right\|^2 + \lambda \sum_{j=1}^{M} \|\mathbf{b}_j\|^2 + \sum_{j=1}^{M} \lambda_{1,j} \|\mathbf{b}_j\|_1$$

Subject to $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_{M \times M}$.

- $\mathbf{A}$: Orthonormal basis (scores).
- $\mathbf{B}$: Sparse loadings vectors $\mathbf{b}_j$.
- ℓ_1 penalty ($\lambda_{1,j}$): Forces small loadings to exactly zero.
- ℓ_2 penalty (λ): Ensures strictly convex problem for $\mathbf{B}$ when $D \gg N$.

ALTERNATING OPTIMIZATION

The objective is not jointly convex in $\mathbf{A}$ and $\mathbf{B}$, but it is convex in each matrix while holding the other fixed. The algorithm alternates:

- **Step 1 (Fix $\mathbf{B}$, update $\mathbf{A}$):** This is a Procrustes rotation problem. Compute SVD of $\mathbf{X}^\top \mathbf{X}\mathbf{B} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, then set $\mathbf{A} = \mathbf{U}\mathbf{V}^\top$.
- **Step 2 (Fix $\mathbf{A}$, update $\mathbf{B}$):** This decouples into M independent Elastic Net regression problems, which can be solved efficiently using LARS (Least Angle Regression).

THE TRADE-OFF

Sparsity vs. Variance Explained. By shrinking loadings to zero, the components become highly interpretable (e.g., identifying specific gene modules), but they are no longer strictly orthogonal, and they explain slightly less total variance than standard PCA.

Probabilistic PCA: A Generative Model for PCA

Tipping and Bishop (1999) showed that PCA can be derived as the maximum likelihood solution of a **latent variable model**:

$$\mathbf{x} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}$$

MODEL COMPONENTS

- $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$: the latent variable
- $\mathbf{W} \in \mathbb{R}^{D \times M}$: factor loading matrix
- $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$: isotropic noise

The marginal distribution is: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^\top + \sigma^2\mathbf{I})$

MAXIMUM LIKELIHOOD & PPCA

As $\sigma^2 \to 0$: The ML estimate of $\mathbf{W}$ spans the same subspace as the top M eigenvectors of $\mathbf{S}$, recovering classical PCA.

For $\sigma^2 > 0$: The model is **Probabilistic PCA (PPCA)**.

Key Advantages

- Enables EM-based inference
- Handles missing data naturally
- Provides a principled framework for model selection

Beyond Linearity: The Need for Kernel PCA

Standard PCA is a **linear** method. It finds the best linear subspace. When the data lies on a **nonlinear manifold**, PCA will fail to capture the true low-dimensional structure.

MOTIVATING EXAMPLES

- **Circle / Sphere:** Intrinsic dim is much lower than ambient.
- **Swiss Roll:** A 2D manifold in 3D; PCA conflates distant points.
- **Concentric Rings:** Not linearly separable in original coordinates.

THE KEY INSIGHT

If we map the data into a high-dimensional (possibly infinite-dimensional) **feature space** and apply PCA there, we can capture nonlinear structure in the original space.

$$\phi : \mathbb{R}^D \to \mathcal{F}$$

The **kernel trick** lets us do this without explicitly computing $\phi(\mathbf{x})$, enabling computations in very high-dimensional feature spaces.

Kernel PCA: Formulation and the Kernel Trick

COVARIANCE IN FEATURE SPACE

In feature space $\mathcal{F}$, the covariance matrix is:

$$\tilde{\mathbf{C}} = \frac{1}{N} \sum_{n=1}^{N} \phi(\mathbf{x}_n)\phi(\mathbf{x}_n)^\top$$

Eigenvectors lie in the span of $\{\phi(\mathbf{x}_n)\}$, so $\mathbf{v} = \sum_n a_n \phi(\mathbf{x}_n)$.

THE KERNEL TRICK

Using the **kernel function** $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}')$, the problem reduces to an $N \times N$ eigenproblem:

$$\mathbf{K}\mathbf{a} = N\lambda\mathbf{a}, \quad K_{nm} = k(\mathbf{x}_n, \mathbf{x}_m)$$

CENTERING IN FEATURE SPACE

We cannot explicitly center in $\mathcal{F}$; instead use the centered kernel matrix:

$$\tilde{\mathbf{K}} = \mathbf{K} - \mathbf{1}_N\mathbf{K} - \mathbf{K}\mathbf{1}_N + \mathbf{1}_N\mathbf{K}\mathbf{1}_N$$

Here $\mathbf{1}_N$ is the matrix with all entries $1/N$.

COMMON KERNELS

KERNEL	FORMULA	GEOMETRY
Linear	$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$	Recovers standard PCA
Polynomial	$k(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^\top \mathbf{x}' + c)^d$	Polynomial feature maps
RBF / Gaussian	$k(\mathbf{x}, \mathbf{x}') = \exp(-\lVert\mathbf{x} - \mathbf{x}'\rVert^2/2\sigma^2)$	Infinite-dimensional feature space

Kernel PCA: Projecting New Test Points

A practical consideration: given a new test point $\mathbf{x}^*$, how do we compute its projection onto the kernel principal components?

The projection onto the m-th kernel principal component is:

$$y_m(\mathbf{x}^*) = \sum_{n=1}^{N} a_{mn}\, k(\mathbf{x}_n, \mathbf{x}^*)$$

Here $\mathbf{a}_m$ is the m-th eigenvector of the centered kernel matrix $\tilde{\mathbf{K}}$, normalized so that $\mathbf{a}_m^\top \mathbf{a}_m = 1/\lambda_m$.

IMPORTANT LIMITATION

Unlike standard PCA, kernel PCA has no straightforward **pre-image**. Reconstructing inputs from kernel coordinates is only approximate and can be costly.

COMPUTATIONAL COMPLEXITY

Kernel PCA requires storing and diagonalizing the $N \times N$ kernel matrix, yielding:

- **Memory:** $O(N^2)$
- **Computation:** $O(N^3)$

For large N, use approximations (Nyström, random features).

PCA vs. Kernel PCA: A Structured Comparison

PROPERTY	STANDARD PCA	KERNEL PCA
Feature map	Identity (linear)	Implicit nonlinear $\phi(\mathbf{x})$
Covariance	$D \times D$ matrix $\mathbf{S}$	$N \times N$ kernel matrix $\mathbf{K}$
Captures	Linear structure	Nonlinear manifold structure
Pre-image	Exact (linear reconstruction)	Approximate only
Scalability	$O(D^3)$ or $O(N^3)$ via dual	$O(N^3)$ — scales with data size
Interpretability	Loadings in original feature space	No direct feature-space interpretation
Kernel choice	N/A	Requires careful selection of k

> Both methods share the same **mathematical backbone**: they solve eigenvalue problems on symmetric positive semi-definite matrices. Kernel PCA is simply PCA applied in a reproducing kernel Hilbert space (RKHS).

ICA: Going Beyond Second-Order Statistics

PCA exploits the **covariance structure** of the data — it finds directions of maximum variance and produces uncorrelated components. However, **uncorrelated does not imply independent** (for non-Gaussian distributions), and PCA components are not unique up to rotation.

Independent Component Analysis (ICA) seeks a linear transformation $\mathbf{x} = \mathbf{A}\mathbf{s}$ such that the **source signals** $\mathbf{s} = (s_1, \ldots, s_M)^\top$ are **statistically independent** — a much stronger condition than uncorrelatedness.

The Cocktail Party Problem

M microphones record M simultaneous speakers. Each microphone records a linear mixture of all voices. ICA recovers the original source signals from the mixtures, using only the assumption that the sources are statistically independent.

KEY CONTRAST WITH PCA

PROPERTY	PCA	ICA
Objective	Maximize variance / minimize reconstruction error	Maximize statistical independence of components
Statistics used	Second-order (covariance)	Higher-order (non-Gaussianity)
Uniqueness	Rotation-invariant (not unique)	Unique up to permutation and scaling
Assumption	Gaussian or arbitrary distribution	Sources must be non-Gaussian

ICA: Formulation and the Role of Non-Gaussianity

THE GENERATIVE MODEL

Assume the observed data $\mathbf{x}$ is a linear mixture of independent latent sources $\mathbf{s}$:

$$\mathbf{x} = \mathbf{A}\mathbf{s}$$

where $\mathbf{A}$ is an unknown mixing matrix. The goal of ICA is to find an unmixing matrix $\mathbf{W} \approx \mathbf{A}^{-1}$ such that the estimated components:

$$\mathbf{y} = \mathbf{W}\mathbf{x}$$

are as statistically independent as possible.

WHY NON-GAUSSIANITY?

By the **Central Limit Theorem**, the distribution of a sum of independent random variables tends toward a Gaussian distribution.

Therefore, a mixture $\mathbf{x} = \mathbf{A}\mathbf{s}$ will be *more Gaussian* than any of its individual non-Gaussian source components s_i.

The Core Principle of ICA: Maximizing the non-Gaussianity of $\mathbf{w}^\top \mathbf{x}$ leads us to the independent components.

If the true sources were Gaussian, their joint distribution would be perfectly spherical (after whitening), making any orthogonal rotation equally valid. **ICA requires at most one Gaussian source.**

ICA: What Can and Cannot Be Recovered

IDENTIFIABILITY

Unlike PCA, the ICA model is **identifiable** (up to certain trivial ambiguities) as long as **at most one source is Gaussian**. The non-Gaussianity of the sources is what breaks the rotational symmetry that plagues PCA.

INHERENT AMBIGUITIES

Because both the mixing matrix $\mathbf{A}$ and the sources $\mathbf{s}$ are unknown in $\mathbf{x} = \mathbf{A}\mathbf{s}$, we cannot determine:

- **Permutation:** The order of the sources is arbitrary. We can swap columns of $\mathbf{A}$ and rows of $\mathbf{s}$ without changing $\mathbf{x}$.
- **Scaling and Sign:** We can multiply a source by a scalar α and divide the corresponding column of $\mathbf{A}$ by α. Thus, we typically fix the variance of the recovered sources to 1.

MODEL LIMITATIONS

- **Linear Mixing:** Standard ICA assumes the observed data are linear combinations of sources. Nonlinear ICA is much harder and generally non-identifiable without strong constraints.
- **Independence:** If the true underlying sources are actually correlated, ICA will force them to be independent, potentially yielding incorrect components.
- **Square Mixing:** Basic ICA assumes the number of sources equals the number of sensors ($M = D$). Overcomplete ($M > D$) and undercomplete ($M < D$) ICA require more complex inference.

Summary: ICA is a powerful tool for blind source separation, but its success hinges critically on the assumptions of **non-Gaussianity** and **statistical independence** of the latent sources.

Lecture Summary: A Unified View

We have explored a family of dimensionality reduction and latent variable models, each relaxing or changing specific assumptions of standard PCA.

METHOD	TYPE	KEY IDEA & OBJECTIVE	PRIMARY USE CASE
Standard PCA	Linear	Maximize variance / minimize reconstruction error	General dimensionality reduction, noise filtering
Sparse PCA	Linear	Add $\ell 1$ penalty for sparse loadings	Interpretable components (e.g., gene modules)
Probabilistic PCA	Linear	Generative latent variable model with Gaussian noise	Handling missing data, Bayesian model selection
Kernel PCA	Nonlinear	PCA in RKHS via the kernel trick	Uncovering nonlinear manifold structures
ICA	Linear	Maximize statistical independence (non-Gaussianity)	Blind source separation (e.g., cocktail party)

The Big Picture: Dimensionality reduction is not just about compression; it is about discovering the underlying **latent structure** that generates the observed data, whether that structure is linear, nonlinear, sparse, or independent.

Next Steps: Advanced Latent Variable Models

While PCA and ICA provide powerful linear foundations, modern machine learning often requires capturing complex, nonlinear manifolds and hierarchical structures. Future lectures will explore:

Factor Analysis

A generalization of Probabilistic PCA where the noise covariance matrix $\mathbf{\Psi}$ is allowed to be diagonal rather than isotropic ($\sigma^2\mathbf{I}$).

KEY FEATURE

Models variable-specific noise (uniqueness), making it scale-invariant and widely used in psychometrics and social sciences.

t-SNE & UMAP

Nonlinear dimensionality reduction techniques specifically designed for embedding high-dimensional data into 2D or 3D spaces for visualization.

KEY FEATURE

Preserves local neighborhood structure (manifold topology) rather than global variance, revealing clusters that PCA misses.

Autoencoders & VAEs

Deep learning architectures that use neural networks to learn highly nonlinear mappings between the data space and the latent space.

KEY FEATURE

Variational Autoencoders (VAEs) provide a fully probabilistic generative framework, extending PPCA to deep, nonlinear hierarchies.

References & Further Reading

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Thank You / Q&A